Analysis of Wholesale Customers' Spending Behaviors

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Introduction

This work aims to explore the *Wholesale Customers Data* dataset through a two-phase approach: clustering and classification. First, an unsupervised analysis will be applied using clustering techniques to identify natural customer groups, without considering the *Channel* labels. This phase will allow us not only to verify whether the observed spending patterns are sufficient to distinguish the existing groups in the dataset, but also to identify the most relevant variables for classification. Subsequently, a classification model will be developed to predict the sales channel (*Channel*), based on the selected spending variables, with the goal of confirming the effectiveness of these variables in distinguishing the identified groups. The dataset was downloaded from the UCI platform: Link to the dataset

Dataset Description

The *Wholesale Customers Data* dataset represents the annual spending of customers of a Portuguese wholesale company and categorizes customers according to two sales channels: "Horeca" (i.e., Hotels, Restaurants, and Cafes) and "Retail" (i.e., retail resellers).
The dataset contains 440 observations and 8 variables, divided into quantitative and categorical:

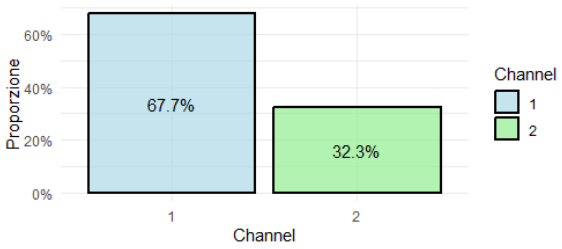
- **Fresh**: Annual spending on fresh products (monetary units, m.u.).
- **Milk**: Annual spending on dairy products (m.u.).
- **Grocery**: Annual spending on packaged food products (m.u.).
- **Frozen**: Annual spending on frozen products (m.u.).
- **Detergents_Paper**: Annual spending on cleaning products and paper (m.u.).
- **Delicassen**: Annual spending on delicatessen products (m.u.).
- **Region**: Customer's region of origin (categorical):
  - 1: Lisbon
  - 2: Oporto
  - 3: Other regions
- **Channel**: Sales channel (categorical):
  - 1: Horeca (Hotels, Restaurants, and Cafes)
  - 2: Retail (Retail sales)

The *Region* variable will not be considered in the analysis, as it is not sufficiently informative for our objectives. The distribution of observations among the categories is unbalanced: only 17.5% of customers come from Lisbon, 10.7% from Oporto, while the majority (71.8%) falls into the generic "Other regions" category. The variable provides limited insight, focusing only on two specific cities (Lisbon and Oporto), which represent a small portion of the entire sample. Moreover, the spending variables are uniformly distributed across the different Region classes, further reinforcing the idea that this variable does not add useful information for the analysis objectives.

Descriptive Statistics

Variable	Min	Max	Mean	Median	SD
Fresh	3	112151	12000.298	8504.0	12647.329
Milk	55	73498	5796.266	3627.0	7380.377
Grocery	3	92780	7951.277	4755.5	9503.163
Frozen	25	60869	3071.932	1526.0	4854.673
Detergents	3	40827	2881.493	816.5	4767.854
Delicassen	3	47943	1524.870	965.5	2820.106

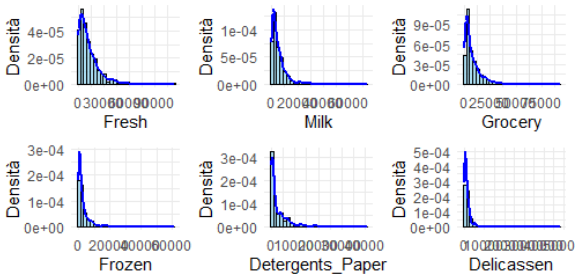
Quantitative Descriptive Statistics



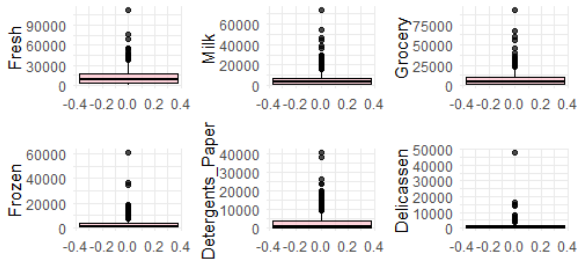
Sales Channel Proportions

Considering we are analyzing sales data, the spending volume among customers is extremely variable, and the descriptive statistics table highlights the wide range of values. Furthermore, it is noted that almost 2/3 of the customers belong to the Horeca channel. The total spending volume for Horeca customers amounts to €7,999,569, while for Retail customers it is €6,619,931.

Dataset Exploration

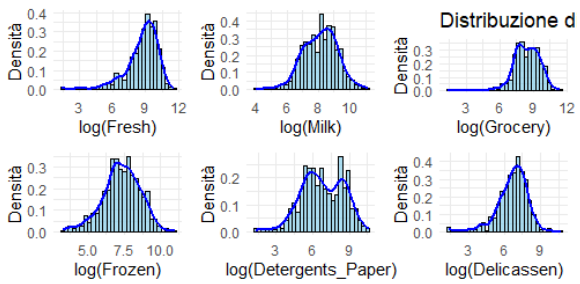


Distributions

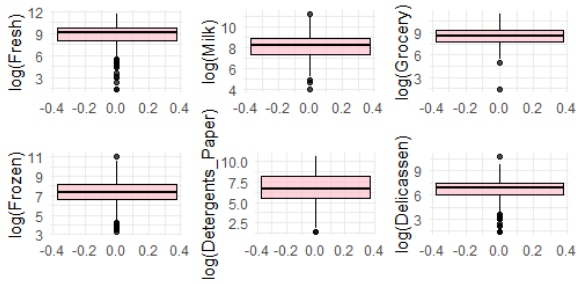


Boxplot

The exploration of the distributions of the spending variables revealed strong skewness, characterized by long tails and a predominance of low values, with a significant presence of outliers, clearly visible in the boxplots. For this reason, a logarithmic transformation of the variables is applied to reduce the impact of outliers and improve distribution symmetry, making the variables more suitable for clustering methods, which could otherwise be negatively influenced by the original skewed structure of the data.

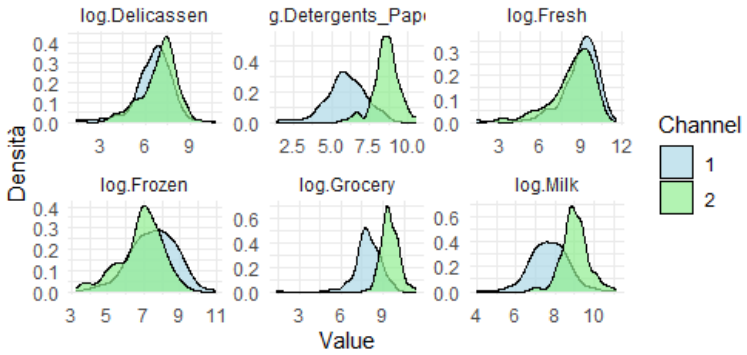


Log Distributions



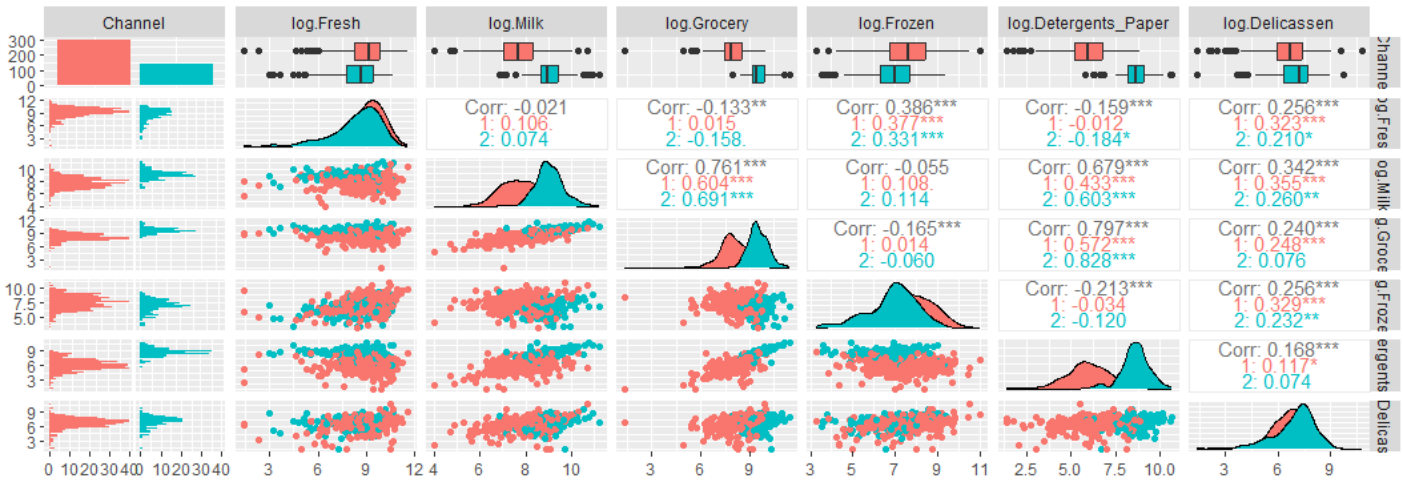
Log Boxplot

After the logarithmic transformation, the variables show more centered distributions with a significant reduction in the number of outliers observed in the boxplots. It is therefore considered appropriate to continue the analysis using the transformed dataset. In particular, some variables such as log(Detergents_Paper), log(Grocery), and log(Milk) appear to exhibit bimodal distributions, suggesting the presence of latent subgroups in the data, which will be further investigated in the subsequent exploratory and clustering analyses.



Log Spending Densities by Channel

The density plot of the log-spending variables, split by Channel, shows significant differences between Retail and Horeca customers in the Detergents_Paper, Grocery, and Milk spending categories. These differences explain the bimodal distribution observed in those variables, confirming that it is attributable to the Channel variable. In particular, Retail customers tend to spend more than Horeca customers in these categories, highlighting distinct spending behaviors between the two sales channels.

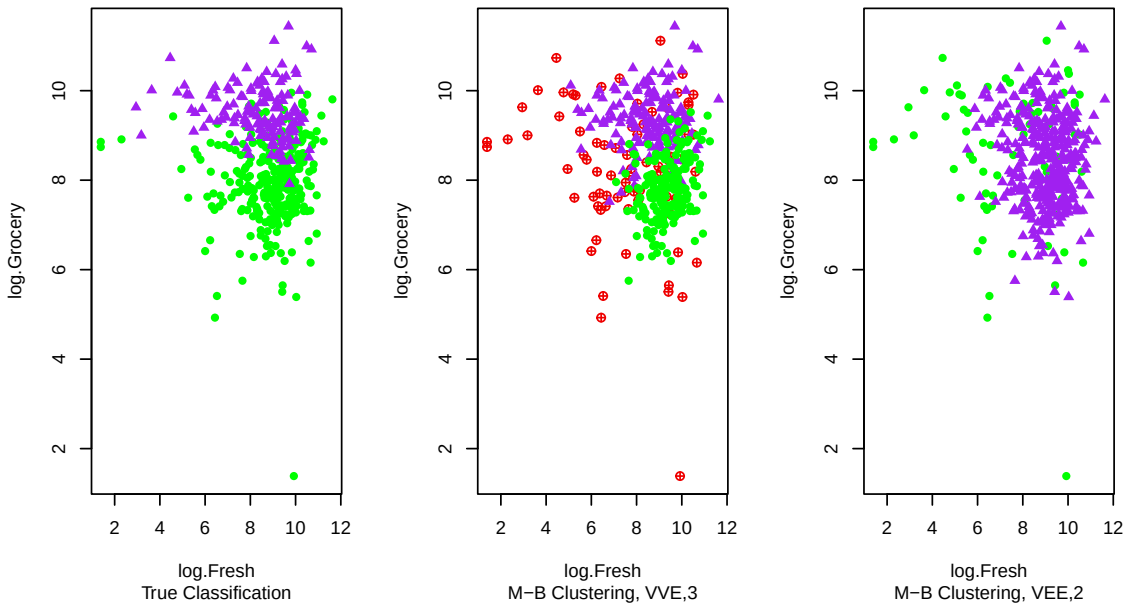


Regarding the spending on Fresh and Frozen products, Horeca customers slightly surpass Retail customers. On the other hand, for Delicatessen products, the spending does not differ significantly between customer groups. Overall, the spending on Milk, Grocery, and Detergents_Paper is positively correlated, as customers tend to purchase these categories jointly, as evidenced by the scatterplots. It is also noted that many Horeca customers display particularly extreme outliers that could pose problems during the classification phase.

Model-Based Cluster

Initially, all variables were considered to identify natural customer groups based solely on spending patterns. A mixture model based on multivariate normal distributions was estimated. Using the BIC criterion, the selected model featured 5 groups, but these did not reflect the Channel labels in any meaningful way, rendering the result insignificant. In contrast, applying the ICL criterion, 3 groups were identified (model VVE,3) which more closely matched the Channel labels. This allowed the selection of a model with better fitting efficiency, balanced complexity, and superior clustering quality. An attempt was also made to force the algorithm to produce two clusters, trying to replicate the natural Channel labels.

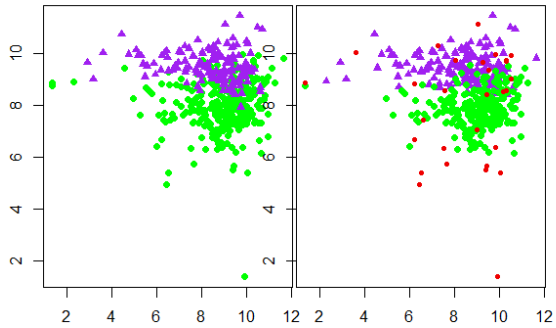
However, the result (model VEE,2) neither aligned with the original labels nor proved useful. The generated division produced one very large group including 82
Even when comparing the indices of the two models, the clear superiority of the VVE model with 3 clusters emerges, showing a CER of 0.284 and an ARI of 0.3599—values indicating greater accuracy and consistency with the dataset's natural labels (Channel). Conversely, the VEE model with 2 clusters shows a much higher CER (0.6045) and a nearly null ARI (0.0104), highlighting its inadequacy in properly representing the data structure.
Below is a chart comparing the true clustering with that obtained from the VVE 3-cluster and VEE 2-cluster models:



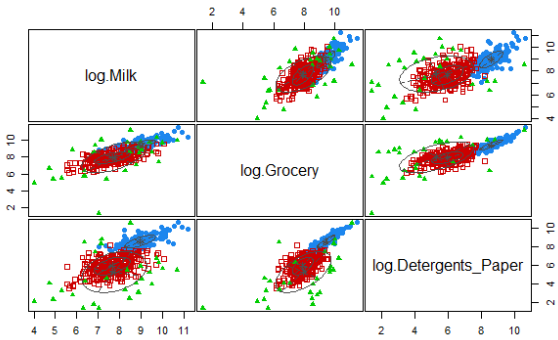
The distribution of groups in the VVE,3 model also appears more coherent visually. However, a problem remains with observations that deviate more strongly from the other groups. This phenomenon may indicate the existence of an additional subclass within Channel, or it may suggest that the high number of variables used makes the model overly complex.

Variable Reduction

Subsequently, the M-B Clustering was re-implemented by removing the three variables that, from the exploratory analysis, did not show significant differences between channels (Fresh, Frozen, and Delicatessen). This led to the selection of the VVV model with 3 clusters. In this configuration, the CER (0.1545) and ARI (0.5470) values indicate a clear improvement compared to the model obtained using all six variables (CER=0.284, ARI=0.3599). This result highlights that removing less informative variables reduced unnecessary model complexity, thereby improving classification quality.



True Classification and M-B-C VVV,3



Classification on all variables

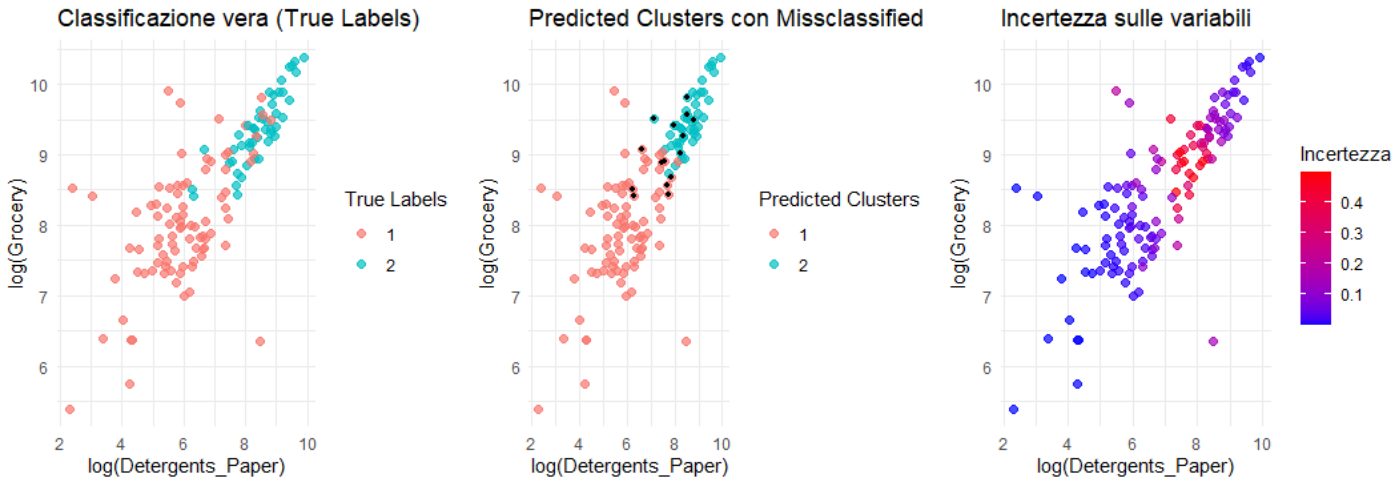
The choice of the VVV model, which allows a fully flexible structure for the variance-covariance matrix, is consistent with the nature of the distributions observed in the different channels. Using only three variables, the model was able to exploit this flexibility to better adapt to the data, avoiding the excessive complexity due to the high dimensionality of the initial six variables. This demonstrates that reducing the dataset's dimensionality—aimed at eliminating less relevant variables—can significantly improve model performance.

Classification

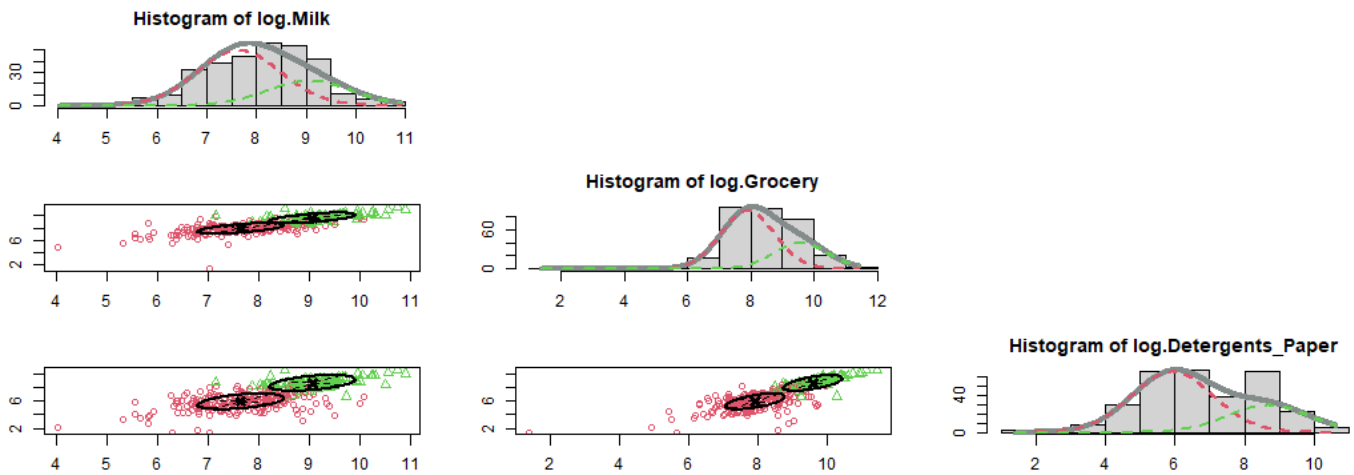
Before proceeding with classification, the dataset was split into a training set, used for model training, and a test set, reserved for evaluating its predictive capabilities, removing the variables previously deemed less relevant during clustering. A classifier based on **EDDA** (Eigenvalue Decomposition Discriminant Analysis) models was then trained, with its performance validated using a cross-validation procedure. Through nested loops, different models were assessed in terms of estimated MER and the optimal number of cross-validation folds.
The best-performing model was **Gaussian_pk_L_C**, characterized by variable volumes between clusters (pk), common shapes (L), and identical orientations (C). This implies that each cluster has a different volume but shares the same shape and orientation in the multidimensional space. The selected model highlights a cluster distribution where group 1 represents 69.16% of observations, characterized by mean logarithmic values of 7.648, 7.909, and 5.926 (€2096, €2721, and €374 spent on average for Milk, Grocery, and Detergents_Paper, respectively), and group 2, which accounts for the remaining 30.84%, with significantly higher means for the various spending components: 9.061, 9.577, and 8.675 (€8612, €14429, and €5854 spent on average for Milk, Grocery, and Detergents_Paper, respectively).
This distribution suggests that the mixture model captures the distinct customer channels. Group 1 (Horeca), the predominant one, represents a clientele with more moderate and likely more homogeneous spending. In contrast, group 2 (Retail) identifies a minority with significantly higher spending, where the demand for quantity and quality of products is generally higher.

Prediction on the Test Set

The classification prediction phase was carried out on the test set using the previously selected model (Gaussian_pk_L_C). The results show a Classification Error Rate of 0.1136 and an ARI of 0.5914, values that indicate excellent model performance in its ability to distinguish the natural groups present in the data.



Observations with greater uncertainty are concentrated in the boundary region between the two clusters, where the cluster distributions overlap the most. This is due to the probabilistic nature of the mixture model, which assigns observations based on their similarity to the estimated distributions. Those near the boundary between clusters have similar probabilities for both groups, making the assignment less precise.



We can conclude that the curves show the mixture model successfully identified well-separated clusters for each variable, capturing both the within-cluster variability and the overlap between them. The selected model, namely a Gaussian mixture with a VVV structure (fully free variances and covariances), proves particularly suitable for capturing the complexity of the data. However, the presence of uncertainty in boundary observations indicates an overlap between clusters, which causes most misclassifications. The structure of the mixture model also suggests that the differences between groups are not only quantitative but likely related to different operational and consumption dynamics, reflecting a natural segmentation of the clientele into the two main channels.